Assignment no. 3

Generative AI
GAI4UE
SS 2026

In this assignment you will train NVIDIA's **pix2pixHD** to translate binary foreground masks into brightfield microscopy images of *C. elegans* worms. The dataset is BBBC010 and the setup mirrors Unit 5 (BBBC039 nuclei), but on a different training set.

Exercise 1 (8 points) Download and prepare BBBC010:

- Fetch the three archives: `BBBC010_v2_images.zip`, `BBBC010_v1_foreground.zip`, and `BBBC010_v1_foreground_eachworm.zip`.
- Pair each well's binary mask with its image (extract the well code from the image filename).
- Binarize masks (any value $> 0 \rightarrow$ foreground), normalise the 16-bit images to 8-bit RGB, and resize everything to **512×512**.
- Split 80% train / 20% test (`random.seed(42)`) and save in pix2pixHD layout (`train_A/train_B/test_A/test_B`).
- Visualise a few train pairs (mask alongside image).

Exercise 2 (2 points) Set up and train pix2pixHD:

- Clone `https://github.com/NVIDIA/pix2pixHD` and apply the `apply_patches.py` script (Python 3.11+ compatibility, checkpoint saving).
- Train with `--label_nc 0 --no_instance --loadSize 512 --fineSize 512 --batchSize 2 --niter 40 --niter_decay 0 --save_epoch_freq 100 --gpu_ids 0`.
- Confirm that checkpoints are saved at epochs **5, 10, 20, 40**.

Exercise 3 (10 points) Evaluate generation quality across epochs:

- Run `test.py` on the test set for each checkpoint.
- Compute **SSIM** (`skimage.metrics.structural_similarity`) between every generated and ground-truth pair.
- Report mean±std SSIM per epoch and plot the trend (scatter + mean line).
- Side-by-side comparison: pick one test well and show its generation at each saved epoch.
- Discuss: do the generated textures look plausible? Show e.g. zoomed in areas? Look for artefacts? Compare with real images?

Skeleton: A starter repository with notebook templates is available at:
`https://github.com/MustafaArikan/GAI4_course` (folder `HW_3`).

Deliverables:

- 3 Jupyter notebooks exported as HTML
- 1-page report (PDF): describe the dataset and pre-processing, summarise the SSIM across epochs, comment on the random-mask experiment, and note any implausible artefacts.

Submission: electronically via Moodle; **Deadline: Mon, Jun 8, 2026, 23:59**